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In [1]: # Meghanatha Reddy Pallem
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# Security Analytics Project-2
# Identify employee accesses to resources outside their own department

import pandas as pd

# Loading the CSV files
access = pd.read_csv("datalogUpdated.csv")
employees = pd.read_csv("employee_details.csv")
resources = pd.read_csv("dept_resources.csv")
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In [2]: # Merging access log with employee details (by EmployeeID)
merged = pd.merge(access, employees, on="EmployeeID", how="left")

# Merging with resource ownership (by ResourceID)
merged = pd.merge(merged, resources, on="ResourceID", how="left")

print("Merged dataset sample:")
print(merged.head())
```

Merged dataset sample:

	Datetime	EmployeeID	Action	ResourceID	Access IPAddress	Latitude \
0	6/1/2023 0:05	10159	login	AuthServer	192.168.1.25	35.175927
1	6/1/2023 0:09	10163	login	AuthServer	192.168.1.49	35.228719
2	6/1/2023 0:10	10162	login	AuthServer	192.168.1.39	35.228719
3	6/1/2023 1:14	10162	logout	AuthServer	192.168.1.39	35.228719
4	6/1/2023 1:25	10163	logout	AuthServer	192.168.1.49	35.228719

	Longitude	Result	Name	Department_x	Start Date	End Date \
0	-80.999608	success	Lenore	Finance	11/28/12	12/30/20
1	-80.821720	success	Maisie	IT	7/15/13	12/30/20
2	-80.821720	success	MacKenzie	Finance	9/27/13	12/30/20
3	-80.821720	success	MacKenzie	Finance	9/27/13	12/30/20
4	-80.821720	success	Maisie	IT	7/15/13	12/30/20

	Home Latitude	Home Longitude	Department_y
0	35.175927	-80.999608	IT
1	35.331766	-80.879161	IT
2	35.274461	-80.689427	IT
3	35.274461	-80.689427	IT
4	35.331766	-80.879161	IT

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In [3]: # Keeping rows where employee's dept != resource's dept
outside = merged[merged["Department_x"] != merged["Department_y"]]

result_df = outside[[
    "EmployeeID",
    "Name",
    "Department_x",
    "ResourceID",
    "Department_y",
    "Action",
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        "Datetime",
        "Result"
    ]].copy()

    # Renaming columns for clarity
    result_df = result_df.rename(columns={
        "Department_x": "Employee_Dept",
        "Department_y": "Resource_Dept"
    })

    print("Filtered sample:")
    print(result_df.head())

```

Filtered sample:

	EmployeeID	Name	Employee_Dept	ResourceID	Resource_Dept	Action	\
0	10159	Lenore	Finance	AuthServer	IT	login	
2	10162	MacKenzie	Finance	AuthServer	IT	login	
3	10162	MacKenzie	Finance	AuthServer	IT	logout	
5	10159	Lenore	Finance	AuthServer	IT	logout	
7	10104	Amanda	Finance	AuthServer	IT	login	

	Datetime	Result
0	6/1/2023 0:05	success
2	6/1/2023 0:10	success
3	6/1/2023 1:14	success
5	6/1/2023 1:56	success
7	6/1/2023 8:09	success

```

In [4]: # Dropping exact duplicate rows
result_df = result_df.drop_duplicates()

# Total count
print(f"Total outside-department access events: {len(result_df)}\n")

# Sample rows
print("First 10 rows:")
print(result_df.head(10))

print("\nLast 10 rows:")
print(result_df.tail(10))

```

Total outside-department access events: 96165

First 10 rows:

	EmployeeID	Name	Employee_Dept	ResourceID	Resource_Dept	Action	\
0	10159	Lenore	Finance	AuthServer	IT	login	
2	10162	MacKenzie	Finance	AuthServer	IT	login	
3	10162	MacKenzie	Finance	AuthServer	IT	logout	
5	10159	Lenore	Finance	AuthServer	IT	logout	
7	10104	Amanda	Finance	AuthServer	IT	login	
9	10174	Quinlan	Finance	AuthServer	IT	login	
11	10116	Bryar	Finance	AuthServer	IT	login	
13	10113	Brenna	Finance	AuthServer	IT	login	
16	10183	Silas	HR	AuthServer	IT	login	
19	10165	Megan	Finance	AuthServer	IT	login	

	Datetime	Result
0	6/1/2023 0:05	success
2	6/1/2023 0:10	success
3	6/1/2023 1:14	success
5	6/1/2023 1:56	success
7	6/1/2023 8:09	success
9	6/1/2023 8:21	success
11	6/1/2023 8:30	success
13	6/1/2023 8:33	success
16	6/1/2023 8:43	success
19	6/1/2023 8:48	success

Last 10 rows:

	EmployeeID	Name	Employee_Dept	ResourceID	Resource_Dept	Action	\
333934	10189	Theodore	Finance	AuthServer	IT	logout	
333936	10162	MacKenzie	Finance	AuthServer	IT	logout	
333940	10162	MacKenzie	Finance	AuthServer	IT	login	
333942	10129	Denise	HR	AuthServer	IT	logout	
333943	10129	Denise	HR	AuthServer	IT	login	
333944	10162	MacKenzie	Finance	AuthServer	IT	logout	
333948	10184	Stacy	HR	AuthServer	IT	logout	
333949	10129	Denise	HR	AuthServer	IT	logout	
333950	10184	Stacy	HR	AuthServer	IT	login	
333951	10184	Stacy	HR	AuthServer	IT	logout	

	Datetime	Result
333934	6/29/2021 21:24	success
333936	6/29/2021 21:33	success
333940	6/29/2021 21:47	success
333942	6/29/2021 22:04	success
333943	6/29/2021 22:15	success
333944	6/29/2021 22:27	success
333948	6/29/2021 23:40	success
333949	6/29/2021 23:47	success
333950	6/29/2021 23:50	success
333951	6/30/2021 0:00	success

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In [5]: # Dropping duplicates by EmployeeID
result_df = result_df.drop_duplicates(subset=["EmployeeID"])

# Total count
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print(f"Total unique employees who accessed outside their department: {len(result_d

#Sample rows
print("First 10 rows:")
print(result_df.head(10))

print("\nLast 10 rows:")
print(result_df.tail(10))
```

Total unique employees who accessed outside their department: 68

First 10 rows:

	EmployeeID	Name	Employee_Dept	ResourceID	Resource_Dept	Action	\
0	10159	Lenore	Finance	AuthServer	IT	login	
2	10162	MacKenzie	Finance	AuthServer	IT	login	
7	10104	Amanda	Finance	AuthServer	IT	login	
9	10174	Quinlan	Finance	AuthServer	IT	login	
11	10116	Bryar	Finance	AuthServer	IT	login	
13	10113	Brenna	Finance	AuthServer	IT	login	
16	10183	Silas	HR	AuthServer	IT	login	
19	10165	Megan	Finance	AuthServer	IT	login	
21	10178	Sacha	HR	AuthServer	IT	login	
22	10192	Wanda	Finance	AuthServer	IT	login	

	Datetime	Result
0	6/1/2023 0:05	success
2	6/1/2023 0:10	success
7	6/1/2023 8:09	success
9	6/1/2023 8:21	success
11	6/1/2023 8:30	success
13	6/1/2023 8:33	success
16	6/1/2023 8:43	success
19	6/1/2023 8:48	success
21	6/1/2023 8:51	success
22	6/1/2023 8:51	success

Last 10 rows:

	EmployeeID	Name	Employee_Dept	ResourceID	Resource_Dept	Action	\
1243	10160	Leo	HR	AuthServer	IT	login	
2065	10118	Callum	Research	AuthServer	IT	login	
2189	10128	Deanna	HR	AuthServer	IT	login	
2197	10187	Susan	Finance	AuthServer	IT	login	
3071	10136	Garrison	HR	AuthServer	IT	login	
3094	10126	Dacey	Finance	AuthServer	IT	login	
3314	10171	Nevada	Finance	AuthServer	IT	login	
5369	10148	Isabella	HR	AuthServer	IT	login	
5892	10152	Jolie	Research	AuthServer	IT	login	
207627	10100	Adara	IT	Server0	HR	access	

	Datetime	Result
1243	6/2/2023 11:17	success
2065	6/3/2023 8:00	success
2189	6/3/2023 10:28	success
2197	6/3/2023 10:30	success
3071	6/4/2023 8:01	success
3094	6/4/2023 8:44	success
3314	6/4/2023 11:26	success
5369	6/6/2023 13:33	success
5892	6/8/2023 8:45	success
207627	1/28/2021 16:43	failed

In []: